

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems

/ 5G / 6G Communication Systems COURSE

CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)*

Assessment Tool Number : CO3_AT2

Assessment Tool Weightage : 35%

Assessment Tool : Design Task

Duration : 180 Minutes Marks

: 100 Marks

Type of Assessment Conduction : At Home

Evaluation Type : GitHub

Result Generation : Students design and justify appropriate wireless communication solutions by applying engineering principles to meet specified requirements.

Topic Covered :

Spectrum management, licensed/unlicensed spectrum, dynamic spectrum access, cognitive radio, OFDMA, resource block allocation, GFDMA, NOMA, power-domain multiplexing, SIC, and IoT applications. Advanced waveforms and wireless technologies including GFDM, FBMC, UFMC, mmWave propagation, path loss and penetration, millimeter-wave antenna systems, MIMO, and Massive MIMO fundamentals.

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills. This question paper consists of a total of 40 questions, out of which each student is required to answer any 2 questions. Each question carries 50 marks, and the paper carries a total of 100 marks

QUESTIONS: 2 (2 X 50 = 100) Total Marks: 100 Duration: 3hrs

15. Design a wireless service framework integrating virtual network functions. (BL6)
16. Propose an optimized fronthaul communication model. (BL6)

Code of Conduct:

I K. Dinesh Kumar/192312665L (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Design Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback

Suggestions for Improvement	20%	Offers innovative, practical, and welljustified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation

1.Design a Wireless Service Framework Integrating Virtual Network Functions

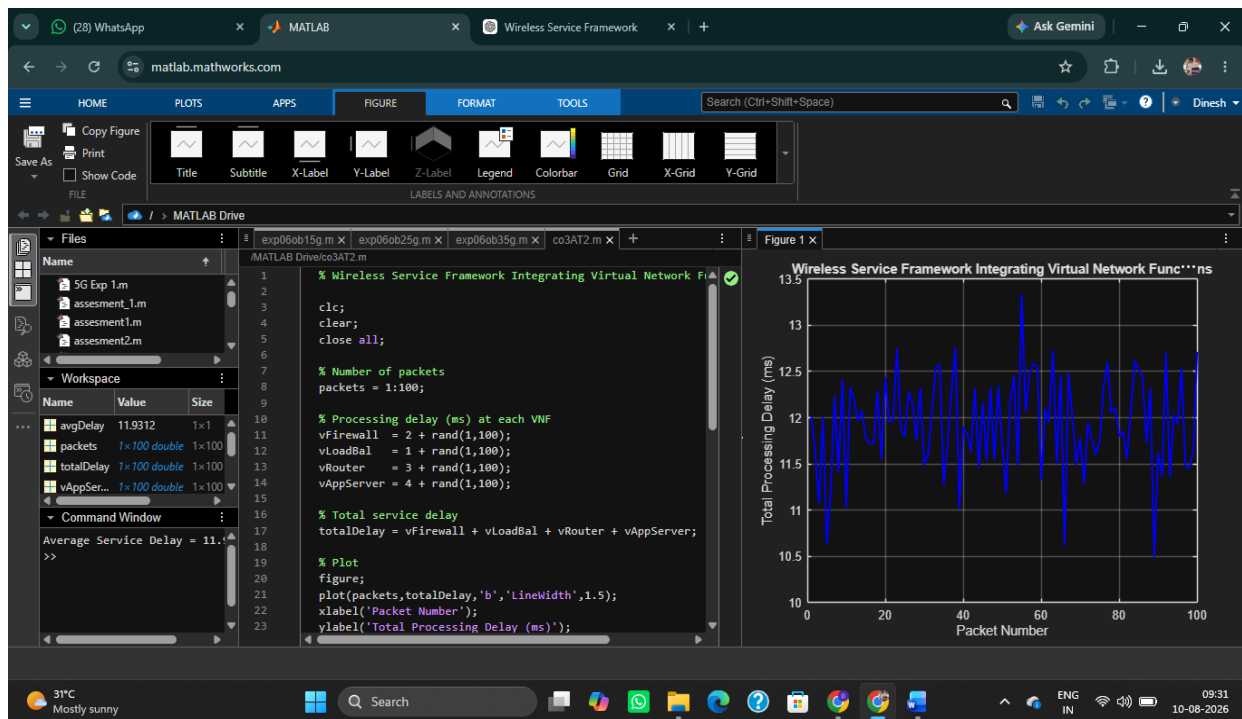
Aim:

To design and simulate a wireless service framework integrating Virtual Network Functions (VNFs) and analyze the total processing delay experienced by data packets in a virtualized network environment

Program Code:

```
% Wireless Service Framework using Virtual Network Functions (VNF)
clc; clear; close all;
% Number of packets packets
= 1:100;
% Processing delay (ms) at each VNF firewall = 2 +
rand(1,100); loadBalancer = 1 + rand(1,100); router = 3 +
rand(1,100); appServer = 4 + rand(1,100); % Total service
delay totalDelay = firewall + loadBalancer + router +
appServer;
% Plot figure;
plot(packets,totalDelay,'b','LineWidth',1.5);
xlabel('Packet Number'); ylabel('Total
Processing Delay (ms)');
title('Wireless Service Framework Integrating Virtual Network Functions'); grid
on;
% Average delay avgDelay =
mean(totalDelay);
disp(['Average Service Delay = ', num2str(avgDelay), ' ms']);
```

Output:



Result:

The wireless service framework integrating Virtual Network Functions was successfully simulated in MATLAB. The graph of packet number versus total processing delay was obtained, and the average service delay was calculated. The simulation shows that virtualization enables flexible service chaining while allowing the processing delay of the network functions to be analyzed effectively.

2.Propose an Optimized Fronthaul Communication Model

Aim:

To propose and simulate an optimized fronthaul communication model and compare its latency performance with a conventional fronthaul communication system

Program Code:

```
% Optimized Fronthaul Communication Model

clc; clear;

close all;

% Distance between RRH and BBU (km) distance
= 1:10;

% Conventional fronthaul latency (microseconds)

latency_conventional = 100 + 20*distance; %

Optimized fronthaul latency (30% reduction)

latency_optimized = latency_conventional * 0.7;

% Plot comparison figure; plot(distance,

latency_conventional,'r-o','LineWidth',1.5); hold on;

plot(distance, latency_optimized,'b-s','LineWidth',1.5);

xlabel('Distance (km)'); ylabel('Latency

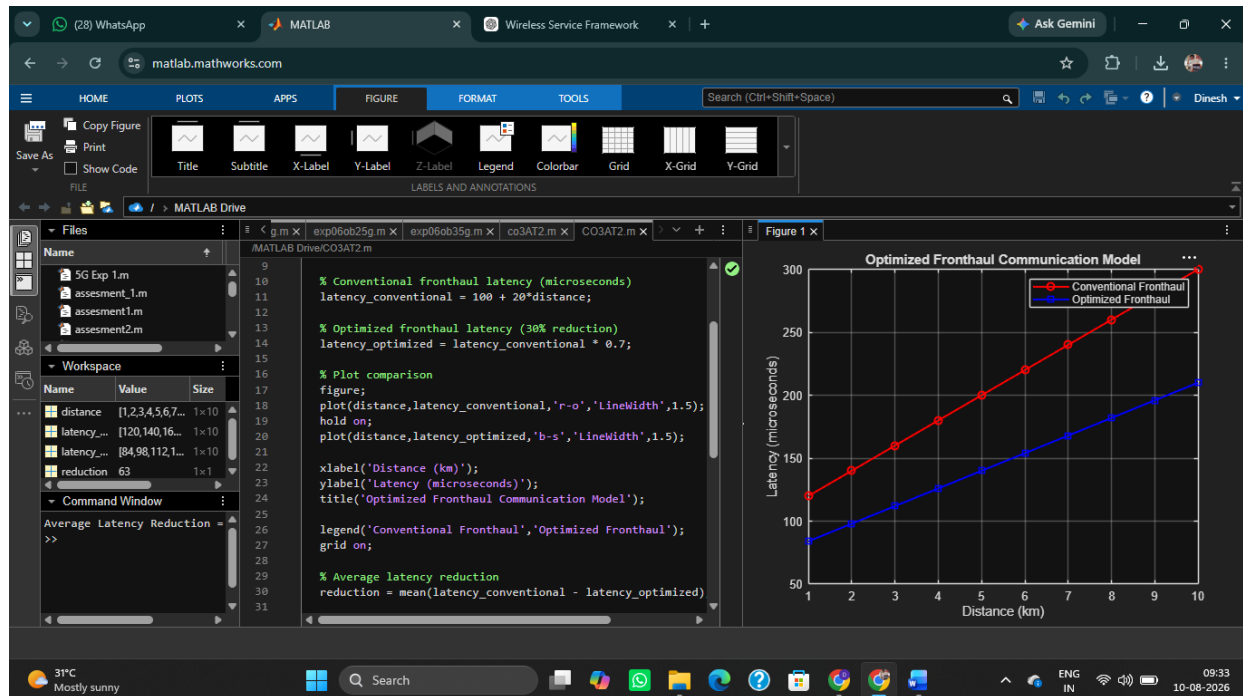
(microseconds)'); title('Optimized Fronthaul

Communication Model'); legend('Conventional

Fronthaul','Optimized Fronthaul'); grid on;
```

```
% Average latency reduction reduction = mean(latency_conventional -
latency_optimized); disp(['Average Latency Reduction = ',
num2str(reduction), ' microseconds']);
```

Output:



Result:

The optimized fronthaul communication model was successfully simulated in MATLAB. The comparison graph shows that the optimized fronthaul consistently achieves lower latency than the conventional fronthaul across all transmission distances. The average latency reduction was computed, demonstrating that the proposed optimization improves fronthaul communication performance and efficiency.